

Assignment-19.1

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Batch – 12

Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

Instructions:

- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

CODE:

```
1 class Student:
2     def __init__(self, name, age):
3         self.name = name
4         self.age = age
5
6     def display(self):
7         print("Name:", self.name)
8         print("Age:", self.age)
9
10 s = Student("Akram", 21)
11 s.display()
```

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Name: Akram
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```

```
1 class Student {
2     Student(String name, int age) {
3         this.name = name;
4         this.age = age;
5     }
6
7     // Method
8     void display() {
9         System.out.println("Name: " + name);
10        System.out.println("Age: " + age);
11    }
12
13    Run | Debug
14    public static void main(String[] args) {
15        Student s = new Student(name: "Akram", age: 21);
16        s.display();
17    }
18 }
```

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\akram\AppData\Roaming\Code\User\workspaceStorage\c81bc8234e12757088efddabec550e35\redhat_java\jdt_ws\AIAC_26af4a4e\bin\' 'Student'
Name: Akram
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```

Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

- Ask AI to translate a recursive C++ factorial function into Python.

- Validate correctness by testing with sample inputs.
- Ensure Python version follows Pythonic conventions.

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

CODE:

```

1 def factorial(n):
2     if n == 0:
3         return 1
4     else:
5         return n * factorial(n-1)
6
7 print(factorial(5))

```

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Name: Akram
Age: 21
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC>

```

```

1 #include <iostream>
2 using namespace std;
3
4 int factorial(int n) {
5     if(n == 0)
6         return 1;
7     else
8         return n * factorial(n-1);
9 }
10
11 int main() {
12     cout << factorial(5);
13     return 0;
14 }

```

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Name: Akram
Age: 21
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC>

```

Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

Instructions:

- Provide AI with a SQL query (e.g., `SELECT name, salary FROM employees WHERE salary > 50000`).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

CODE:

The screenshot shows the Visual Studio Code interface with a Python file named `task_3.py` open. The code defines a dictionary `data` with names and salaries, creates a `pandas` `DataFrame`, and filters for salaries greater than 50,000. The terminal shows the command `python task_3.py` being executed, resulting in the following output:

```
Age: 21
name salary
1 Alice 60000
2 Bob 75000
```

The screenshot shows the Visual Studio Code interface with a SQL file named `task_3.sql` open. The code contains a SQL query to select names and salaries from an `employees` table where the salary is greater than 50,000. The terminal shows the command `python task_3.py` being executed, resulting in the following output:

```
name salary
1 Alice 60000
2 Bob 75000
```

Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

- Converting a Python web scraping snippet into JavaScript for browser automation.

- Translating a Java method into C# for enterprise applications.

Instructions:

- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.

- Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

- Working equivalent code snippets in at least two different programming languages

CODE:

The screenshot shows a Visual Studio Code editor window with a project named 'AIAC'. The Explorer sidebar on the left shows a file structure with folders like 'SOURCE CONTROL CHANGES', 'AIAC', and 'pycache_', and files like 'launch.json', 'tasks.json', 'task_1.java', 'task_1.py', 'task_2.c++', 'task_3.py', 'task_3.sql', 'task_4.py', and 'task2.py'. The main editor area displays the content of 'task_4.py', which is a Python script using the 'requests' and 'BeautifulSoup' libraries to fetch data from 'https://www.onlinegdb.com/'. The script includes comments and a link to a GitHub repository. The bottom panel shows the 'TERMINAL' tab with the command 'python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_4.py' executed successfully, outputting the content of the linked page. The status bar at the bottom indicates the file is encoded in UTF-8, has 4 spaces, and is using the Python 3.13 (64-bit) interpreter.

```

1 import requests
2 from bs4 import BeautifulSoup
3
4 url = "https://www.onlinegdb.com/"
5
6 response = requests.get(url)
7 soup = BeautifulSoup(response.content, 'html.parser')
8
9 for link in soup.find_all('a'):
10     print(link.get("href"))

```

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_4.py
#
#tab-stdin
#tab-stdout
#tab-stderr
#tab-debug-console
/upgrade-premium
https://www.onlinegdb.com/blog/brief-guide-on-how-to-use-onlinegdb-debugger/
#debug_window_call_stack
#debug_window_local_variables
#debug_window_registers
#debug_window_display_expression
#debug_window_breakpoints
#
#

```

